

NATIONAL UNIVERSITY OF SINGAPORE

CS5340 – Uncertainty Modeling in AI

(Semester 1 : AY2025/26)

Time Allowed : 2 Hours

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FOUR** questions and comprises **SEVEN** printed pages.
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is a **CLOSED BOOK** assessment.
6. Students can bring in **ONE A4 summary sheet**.
7. Non-programmable electronic calculators are allowed.

Question 1

Consider the following hierarchical Bayesian model with a latent class variable $Z_n \in \{1, 2\}$ for each data pair (x_n, y_n) , and let $\mathbf{z}_n = (z_{n1}, z_{n2})$ be its one-hot encoding with $z_{nk} \in \{0, 1\}$ and $\sum_{k=1}^2 z_{nk} = 1$:

$$p(\mathbf{z}_n | \boldsymbol{\pi}) = \prod_{k=1}^2 \pi_k^{z_{nk}}, \quad \sum_{k=1}^2 \pi_k = 1, \quad 0 \leq \pi_k \leq 1,$$

$$p(x_n | z_{nk} = 1) = \mathcal{N}(x_n; \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_n - \mu_k)^2}{2\sigma_k^2}\right),$$

$$p(y_n | x_n, z_{nk} = 1) = \mathcal{N}(y_n; a_k x_n + b_k, \tau^2) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(y_n - (a_k x_n + b_k))^2}{2\tau^2}\right).$$

All N data points are independent and identically distributed (*i.i.d.*) according to the parameters: $\theta = \{\pi_k, \mu_k, \sigma_k^2, a_k, b_k, \tau^2\}$. We estimate θ using the Expectation–Maximization (EM) algorithm.

- i) Derive the posterior responsibility $\gamma(z_{nk})$ in the E-step. (5 marks)
- ii) Using the result from the E-step, derive the update equation for each parameter in the M-step. (12 marks)
- iii) Given $N = 3$ data points: $(x_n, y_n) = (1.0, 2.1), (2.0, 3.9), (3.0, 6.1)$ with current parameters: $\pi_1^{(t)} = 0.6, \pi_2^{(t)} = 0.4, \mu_1^{(t)} = 1, \mu_2^{(t)} = 3, \sigma_1^{2(t)} = \sigma_2^{2(t)} = 1, (a_1^{(t)}, b_1^{(t)}) = (2.0, 0), (a_2^{(t)}, b_2^{(t)}) = (1, 1), \tau^{2(t)} = 0.25$.

Compute the responsibilities $\gamma(z_{nk})$ for each data point numerically (3 decimal places).

(3 marks)

- iv) Briefly explain how maximizing Evidence Lower Bound (ELBO) $\mathcal{L}(q, \theta)$ alternately over q and θ leads to the EM updates derived in i) and ii). (5 marks)

Show all your workings clearly.

Useful Equations:

1. $\frac{\partial \ln f(x)}{\partial x} = \frac{1}{f(x)} \frac{\partial f(x)}{\partial x}.$

2. $\frac{\partial \exp f(x)}{\partial x} = \frac{\partial f(x)}{\partial x} \exp f(x).$

3. The solution to a linear system of equations: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ is:

$$x_1 = \frac{b_1 a_{22} - a_{12} b_2}{a_{11} a_{22} - a_{12} a_{21}}, \quad x_2 = \frac{a_{11} b_2 - b_1 a_{21}}{a_{11} a_{22} - a_{12} a_{21}}, \quad \text{given that } a_{11} a_{22} - a_{12} a_{21} \neq 0.$$

Question 2

- i) Consider a Bayesian network consisting of three binary variables arranged in a directed chain $A \rightarrow B \rightarrow C$, where $A, B, C \in \{0, 1\}$. The network is trained from $N = 6$ fully observed data samples as shown below:

Sample #	1	2	3	4	5	6
A	0	0	1	1	1	1
B	0	1	0	1	1	0
C	0	1	0	1	0	0

Let us denote: $\theta_A = p(A = 1)$, $\theta_{B|A=a} = p(B = 1 \mid A = a)$, $\theta_{C|B=b} = p(C = 1 \mid B = b)$, where $a, b \in \{0, 1\}$. The full parameter set of the network is thus: $\theta = \{\theta_A, \theta_{B|A=0}, \theta_{B|A=1}, \theta_{C|B=0}, \theta_{C|B=1}\}$.

- a) Derive the MLE estimates of θ and compute the numerical values.

(6 marks)

- b) Suppose we impose Beta priors $p(\theta \mid \alpha, \beta) = \text{Beta}(\alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$, where $\alpha, \beta > 0$ are the hyperparameters given by:

$$\theta_A \sim \text{Beta}(2, 2), \quad \theta_{B|A=0}, \theta_{B|A=1} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(3, 2), \quad \theta_{C|B=0}, \theta_{C|B=1} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(1, 3).$$

Derive the MAP estimates of θ and compute the numerical values. **(12 marks)**

- c) Comment briefly on how the MAP estimates differ from the MLE in this example and explain why the difference arises. **(1 mark)**

- ii) Now suppose variable C from (i) is connected to a continuous variable $Y \in \mathbb{R}$ through an undirected edge forming a hybrid network: $A \rightarrow B \rightarrow C - Y$. Conditional on C , the variable Y follows a Gaussian: $p(Y \mid C = c) = \mathcal{N}(\mu_c, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu_c)^2}{2\sigma^2}\right)$, where $\sigma^2 = 1$. We observe the following continuous values corresponding to the six samples above:

Sample #	1	2	3	4	5	6
C	0	1	0	1	0	0
Y	0.1	2.2	-0.3	2.0	0.2	0.0

- a) Derive the MLE estimates $\mu_{c=0}$ and $\mu_{c=1}$, and compute the numerical values.

(4 marks)

- b) Using the MLE estimates from part (a), derive the predictive distribution for a new observation Y_{new} given each class label $C = c$. Express $p(Y_{\text{new}} \mid C = c)$ explicitly, and provide the numerical values of the predicted mean and variance for each class.

(2 marks)

Show all your workings clearly.

Useful Equations:

$$1. \quad \frac{\partial \ln f(x)}{\partial x} = \frac{1}{f(x)} \frac{\partial f(x)}{\partial x}.$$

$$2. \quad \frac{\partial \exp f(x)}{\partial x} = \frac{\partial f(x)}{\partial x} \exp f(x).$$

Question 3

A renewable-energy company models the daily solar-panel power output Y_t (in kWh) based on two measured covariates: Ambient temperature T_t ($^{\circ}\text{C}$) and Solar irradiance I_t (W/m^2). The company assumes the following linear-Gaussian model for each day $t = 1, \dots, 4$: $Y_t = \alpha T_t + \beta I_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, where α and β are regression coefficients, and σ^2 represents noise variance. In matrix form, $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \sigma^2) = \mathcal{N}(\mathbf{y} \mid \mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I}_N)$, where

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} T_1 & I_1 \\ T_2 & I_2 \\ T_3 & I_3 \\ T_4 & I_4 \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

$\mathbf{I}_N$ is an $N \times N$ identity matrix, and N is the number of data points. The priors are: $p(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_0)$, where $\boldsymbol{\Sigma}_0 = 10^2 \mathbf{I}_2$, and $p(\sigma^2) = \text{Inv-Gamma}(a_0, b_0)$, where $a_0 = b_0 = 2$. The dataset is:

Day t	1	2	3	4
T_t ($^{\circ}\text{C}$)	28	32	30	29
I_t (W/m^2)	400	600	500	450
Y_t (kWh)	18.5	22.0	19.2	19.7

- i) Assume $\sigma^2 = 1$ is *known*. The current state of the remaining parameters is $\boldsymbol{\theta} = (0.40, 0.008)^{\top}$, and a proposal $\boldsymbol{\theta}' = (0.60, 0.012)^{\top}$ is drawn from a symmetric Gaussian proposal $q(\boldsymbol{\theta}' \mid \boldsymbol{\theta}) = q(\boldsymbol{\theta} \mid \boldsymbol{\theta}')$.
 - a) Derive the expression of the Metropolis-Hastings acceptance probability $\mathcal{A}$. (5 marks)
 - b) Compute numerically the residual sums of squares (RSS): $\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$ and $\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}'\|_2^2$ using the data table. (5 marks)
 - c) With $\boldsymbol{\Sigma}_0^{-1} = 0.01 \mathbf{I}_2$, compute the log-posterior difference: $\Delta = \log p(\boldsymbol{\theta}' \mid \mathbf{y}) - \log p(\boldsymbol{\theta} \mid \mathbf{y})$, and the acceptance probability $\mathcal{A}$. State whether to accept or reject. (5 marks)
- ii) We now treat σ^2 as *unknown* while keeping the same Gaussian likelihood $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \sigma^2)$ and prior $p(\sigma^2)$ with $a_0 = b_0 = 2$.
 - a) We can draw a sample from the conditional posterior of σ^2 given $\mathbf{X}$, $\boldsymbol{\theta}$ and $\mathbf{y}$ using Gibbs sampling. Derive this conditional posterior and show that it follows an Inverse-Gamma distribution.
(Hint: Ignore terms independent of σ^2 to simplify the derivation.) (6 marks)
 - b) Given a sample $u \sim \text{Uniform}(0, 1)$ drawn from a uniform distribution, describe the steps required to draw a sample of σ^2 from its conditional posterior distribution obtained in Part (a). (4 marks)

Show all your workings clearly.

Useful Equations:

1. Inverse-Gamma Distribution $\text{Inv-Gamma}(a, b)$:

$$p(x \mid a, b) = \frac{b^a}{\Gamma(a)} x^{-(a+1)} \exp\left(-\frac{b}{x}\right), \quad x > 0, \text{ where } \Gamma(2) = 1.$$

2. Multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

$$p(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^d.$$

3. Determinant of a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $|A| = ad - bc$.

4. Linear-Gaussian Distribution in matrix form $\mathcal{N}(\mathbf{y} \mid \mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I}_N)$:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left(-\frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2}{2\sigma^2}\right), \text{ where } \|\cdot\|_2^2 \text{ denotes the sum of squared elements.}$$

Question 4

A binary image has N pixels on a 2-D grid. Each pixel i has a latent clean label $X_i \in \{-1, +1\}$ and an observed noisy label $Y_i \in \{-1, +1\}$. The joint distribution is:

$$p(\mathbf{X}, \mathbf{Y}) = \frac{1}{Z_0} \exp\left(\sum_i L_i(X_i) + \sum_{(i,j) \in \mathcal{E}} W_{ij} X_i X_j\right),$$

where $\mathcal{E}$ is the set of neighbor pairs, $W_{ij} > 0$ encourages neighboring pixels to agree, $L_i(X_i)$ is a site-wise log-likelihood term determined by Y_i , and Z_0 is the partition function.

i) Explain why exact inference of $p(\mathbf{X} \mid \mathbf{Y})$ is computationally intractable on large grids. What structural assumption does the mean-field variational approximation make to enable tractable inference? **(2 marks)**

ii) Assume the mean-field factorization $q(\mathbf{X}) = \prod_i q_i(X_i)$ and define $\mu_i = \mathbb{E}_q[X_i]$. Derive the log of the optimized factor $\log q_i^*(X_i)$ up to an additive constant in terms of the neighbor means $\{\mu_j\}_{j \in \text{nbr}(i)}$ and log-likelihood. Clearly indicate which terms depend on X_i . **(4 marks)**

iii) Use the sigmoid parameterization:

$$q_i(+1) = \sigma(2a_i), \quad q_i(-1) = \sigma(-2a_i), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Starting from your result in Part (ii):

- Derive a_i in terms of $L_i(+1)$, $L_i(-1)$, and mean-field influence $m_i = \sum_{j \in \text{nbr}(i)} W_{ij} \mu_j$.
- Derive a closed-form fixed-point update for the new mean μ_i at pixel i .

(7 marks)

iv) Discuss qualitatively how increasing W_{ij} affects: a) convergence behavior of the mean-field iterations, and b) the smoothness of the final denoised labels.

(2 marks)

v) Consider a 2×2 grid with 4-neighborhoods and uniform $W_{ij} = 0.6$. Let $L_i(X_i) = \beta X_i Y_i$ with $\beta = 1.0$ and

$$\begin{bmatrix} Y_{(0,0)} & Y_{(0,1)} \\ Y_{(1,0)} & Y_{(1,1)} \end{bmatrix} = \begin{bmatrix} +1 & -1 \\ -1 & +1 \end{bmatrix}.$$

Initialize $\mu_i^{(0)} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and perform two synchronous mean-field updates: a) Compute $\mu^{(1)}$ and $\mu^{(2)}$ to 3 decimal places. b) Report MAP labels $\hat{X}_i = \text{sign}(\mu_i^{(2)})$.

(10 marks)

Show all your workings clearly.